Tactix: Proof-Checker Classification and Sweep Gate

Report to Andrew

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Outcome

The AM427 Sweep state-machine candidate remains a source-parity candidate, not N4 acceptance. Its matched Tier A comparison is terminal and checker-clean, but has no material performance gain. A new failure in the optional forward every-lemma audit was isolated to a previously documented, uncommitted local `dsr-trim` optimization, not the Tactix proof. The canonical no-option checker contract and the separate H2 failure are unchanged. The AM428 technical report holds exact hashes and commands.

Evidence and decision

The AM427 and parent Tier A arms each solved 199/200 instances: 82 SAT and 117 UNSAT, with the same UNKNOWN. All 117 DSRs per arm independently checked; all per-instance verdicts and proof sizes matched. The candidate took three more solver seconds in aggregate across 200 cases (PAR-2 1.09 versus 1.07 seconds). This is not evidence of an improvement or a broad exercise of the new Sweep branch. The array was `1722828`, aggregate `1722829`, at 4 physical cores/8 CPUs and 32 GiB per task, unthrottled.

On fixed corpus-034, the candidate's exact DSR verifies in both canonical and forward modes when checked with the frozen AM369 `dsr-trim` binary. The ASan `micro_proofs` suite passes 1/1, including negative controls. The shared modified checker clone was restored to its pre-diagnostic binary hash without replacing its owner's edits. The benchmark runner now accepts explicit native-DSR checker pins, records their hashes in its manifest, and checks those bytes before each task; its 29 proof-oracle unit tests pass.

Matched Tier B array `1723236` is in progress on the immutable AM427 and parent binaries with frozen checker pins. A Tier C array was submitted while B owned those build directories; its lock request was refused and the entire still-queued C array and aggregate were cancelled. C will be resubmitted only after B releases the paths. No Tier D run has started.

Open gates and next actions

The H2 native-DSR proof remains rejected by the frozen no-option checker; an option-based acceptance change has not been authorized or validated as a replacement. Broad N1/N2 fuzz, N3 performance, remaining N4 scheduling and reference-trace checks are open. Online pass-order/dose control is suitable for opt-in prototyping, but production learning still needs a shared default-exact slot controller and owned safe-point switching for branching and full clause-deletion sub-strategies. Finish B, repair the runner's lock-acquisition failure path, resubmit matched C, and continue the H2 and fuzz gates. Submit the requested full 5,000-second Tier D comparison only after the goal's correctness and scheduling gates are actually satisfied.